

Emily Lei Kang | Curriculum Vitae

Department of Statistics and Data Science
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Education

The Ohio State University <i>PhD in Statistics</i>	Columbus, OH 2004–2009
Tianjin University <i>BS in Applied Mathematics</i>	Tianjin, China 2000–2004
Nankai University <i>BA in Finance</i>	Tianjin, China 2001–2004

Research Interests

- Statistical methodologies and computational statistics for large data with complex dependence structures, in particular spatial, spatio-temporal and network data
- Uncertainty quantification, parameter calibration and statistical emulation for computer experiments
- Bayesian methodologies and modeling with applications in environmental, climate, biological sciences and engineering
- Statistical learning methods and algorithms for high-dimensional data and variable selection

Professional Experience

Department of Statistics and Data Science <i>Professor</i>	University of Cincinnati 2026–present
Department of Mathematical Sciences <i>Professor</i>	University of Cincinnati 2021–2026
Department of Mathematical Sciences <i>Associate Professor</i>	University of Cincinnati 2016–2021
Statistical and Applied Mathematical Sciences Institute (SAMSI) <i>Visiting Research Fellow</i>	Research Triangle Park, NC 2017 Fall
Department of Mathematical Sciences <i>Assistant Professor</i>	University of Cincinnati 2011–2016

Awards and Honors

2025–: Graduate Fellow (elected), University of Cincinnati
2019: Early Investigator Award from the American Statistical Association Section on Statistics and the Environment
2017,2018: Nominated for the George Barbour Award for Good Faculty-Student Relations, University of Cincinnati
2013: Travel Award from the Association of Women in Mathematics (AWM)-National Science Foundation (NSF)
2012: Faculty Research Fellowship from the University Research Council (URC) at University of Cincinnati

Selected Publications (* denotes a current/former student co-author.)

1. Gyamfi, E. H.*, Konomi, B. A., Lin, G., and **Kang, E. L.** (2026) Enhancing Gaussian Processes for Surrogate Modeling: A Review of Dimension Reduction Techniques for Input Variables. In *Handbook of Statistical Methods for Computer Models: Uncertainty Quantification*, in press.
2. Ekanayaka, A.*, **Kang, E. L.**, Braverman, A., and Kalmus, P. (2025) A multivariate spatial statistical model for statistical downscaling of sea surface temperature in the Great Barrier Reef region. *Journal of the Royal Statistical Society Series C: Applied Statistics*, 74(4), 1183–1213, <https://doi.org/10.1093/jrssc/qlaf019>.
3. Ren, S.*, Wang, P. and **Kang, E.L.** (2025) A Practical Tool for Visualizing and Measuring Model Selection Uncertainty, *Stat*, in press.
4. Cheng, S.*, Konomi, B. A., Karagiannis, G., and **Kang, E. L.** (2024) Recursive nearest neighbor co-kriging models

for big multi-fidelity spatial data sets. *Environmetrics*, <https://doi.org/10.1002/env.2844>.

5. Konomi, B. A., **Kang, E. L.**, Almomani, A.*, and Hobbs, J. (2023) Bayesian Latent Variable Co-kriging Model in Remote Sensing for Quality Flagged Observations, *Journal of Agricultural, Biological and Environmental Statistics*, 28, 423-441.
6. Yang, B., Liu, H., **Kang, E. L.**, Hawthorne, T. L., Tong, S.T.Y., Song, S., and Xu, M. (2022) Traffic restrictions during the 2008 Olympic Games reduced urban heat intensity and extent in Beijing. *Communications Earth & Environment*, 3, 105, <https://doi.org/10.1038/s43247-022-00427-4>.
7. Cheng, S.*, Konomi, B. A., Matthews, J. L., Karagiannis, G., and **Kang, E. L.** (2021) Hierarchical Bayesian nearest neighbor co-kriging Gaussian process models; an application to intersatellite calibration. *Spatial Statistics*, 44, <https://doi.org/10.1016/j.spasta.2021.100516>.
8. Ma, P.*, Mondal, A., Konomi, B. A., Hobbs, J., Song, J. J. and **Kang, E. L.** (2021) Computer model emulation with high-dimensional functional output in large-scale observing system uncertainty experiments, *Technometrics*, 64, 65–79.
9. Cawse-Nicholson, K., Braverman, A., **Kang, E. L.**, Li, M.*, Johnson, M., Halverson, G., Anderson, M., Hain, C., Gunson, M., and Hook, S. (2020) Sensitivity and uncertainty quantification for the ECOSTRESS evapotranspiration algorithm - DisALEXI, *International Journal of Applied Earth Observation and Geoinformation*, <https://doi.org/10.1016/j.jag.2020.102088>
10. Ma, P.* and **Kang, E. L.** (2020) A fused Gaussian process for very large spatial data, *Journal of Computational and Graphical Statistics*, 29, 479-489. [Early version wins the Student Paper Competition in the ICSA Applied Statistics Symposium]
11. Ren, S.*, **Kang, E. L.**, and Lu, J. L. (2020) MCEN: a method of simultaneous variable selection and clustering for high dimensional multinomial regression, *Statistics and Computing*, 30, 291-304.
12. Li, M.*, and **Kang, E. L.** (2019) Randomized algorithms of maximum likelihood estimation with spatial autoregressive models for large-scale networks. *Statistics and Computing*, 28, 1165-1179. [Winner of 2019 student paper competition of ASA Business & Economic Statistics Section]
13. Ma, P.*, **Kang, E. L.**, Braverman, A., and Nguyen, H. (2019) Spatial statistical downscaling for constructing high-resolution nature runs in global observing system simulation experiments. *Technometrics*, 61, 322-340, <https://doi.org/10.1080/00401706.2018.1524791>.

Completed External Funding

- NSF (PI, with G. Lin, Purdue University and B. A. Konomi, University of Cincinnati, as Co-PIs): Collaborative Research: Inference and Uncertainty Quantification for High Dimensional Systems in Remote Sensing: Methods, Computation, and Applications, 2021-2025.
- NASA (Co-I): Neighborhood-scale extreme humid heat health impacts, 2022-2025.
- NASA (Co-I): Ecological Projection Analytic Collaborative Framework (EcoPro), 2022-2025.

Selected Service

- Chair for Committee for 2026 ISBA EnviBayes Student Paper Competition, 2025-2026
- Chair, ISBA EnviBayes, 2025 (Chair-Elect, 2024)
- Committee member, Advanced Research Computing (ARC) Advisory Council, University of Cincinnati, 2025
- Committee Chair for the 2025 Student Paper Competition, Section on Statistics and the Environment, American Statistical Association, 2024-2025
- Associate Editor, *Statistics and Data Science in Imaging*, 2024-present
- Associate Editor, *SIAM/ASA Journal on Uncertainty Quantification (JUQ)*, 2024-present
- Editorial Board Member, *Spatial Statistics*, 2021-present
- Associate Editor, *Journal of Statistical Computation and Simulation*, 2013-present
- Conference Program Committee Member, SIAM UQ 2024
- Committee member, American Statistical Association Committee on Climate Change Policy, 2021-2026
- Program Committee member and North America Representative, Caucus for Women in Statistics, 2022-2023
- Program Chair, Section on Statistics and the Environment, American Statistical Association, 2022
- Associate Editor, *Computers & Geosciences*, 2019-2023
- Core Team Member, US Climate Variability and Predictability (CLIVAR) Data Science Working Group, 2019-2022
- Associate Editor, *Technometrics*, 2022-2023
- Proposal and panel reviewer for NASA, NIH, and NSF